Breast Cancer Mortality Disparities in Michigan

Where Outcomes Differ and Where to Act

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Mortality Varies Despite High Screening

- **Mortality varies widely across counties**

  Significant differences in mortality persist even in counties with similar screening rates

- **Screening rates remain consistently high**

  Screening rates across regions are tightly clustered between 75% and 82%

⚠ Improving screening alone will not reduce mortality disparities

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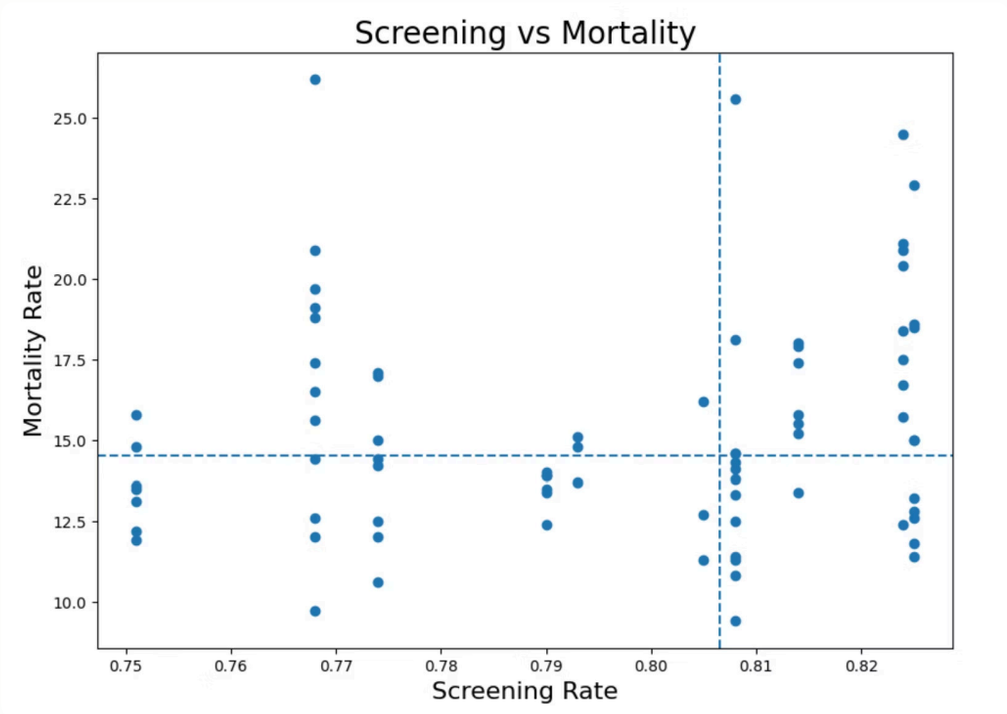
Mortality Varies Despite Consistently High Screening



Screening rates are measured at the regional level; mortality is measured at the county level

Screening rates are similar

Regional screening rates show little variation across the state

Outcomes vary significantly

County-level mortality varies widely, with some counties experiencing 2–3× higher rates

Similar screening, very different outcomes

Differences in mortality are driven by factors beyond screening access

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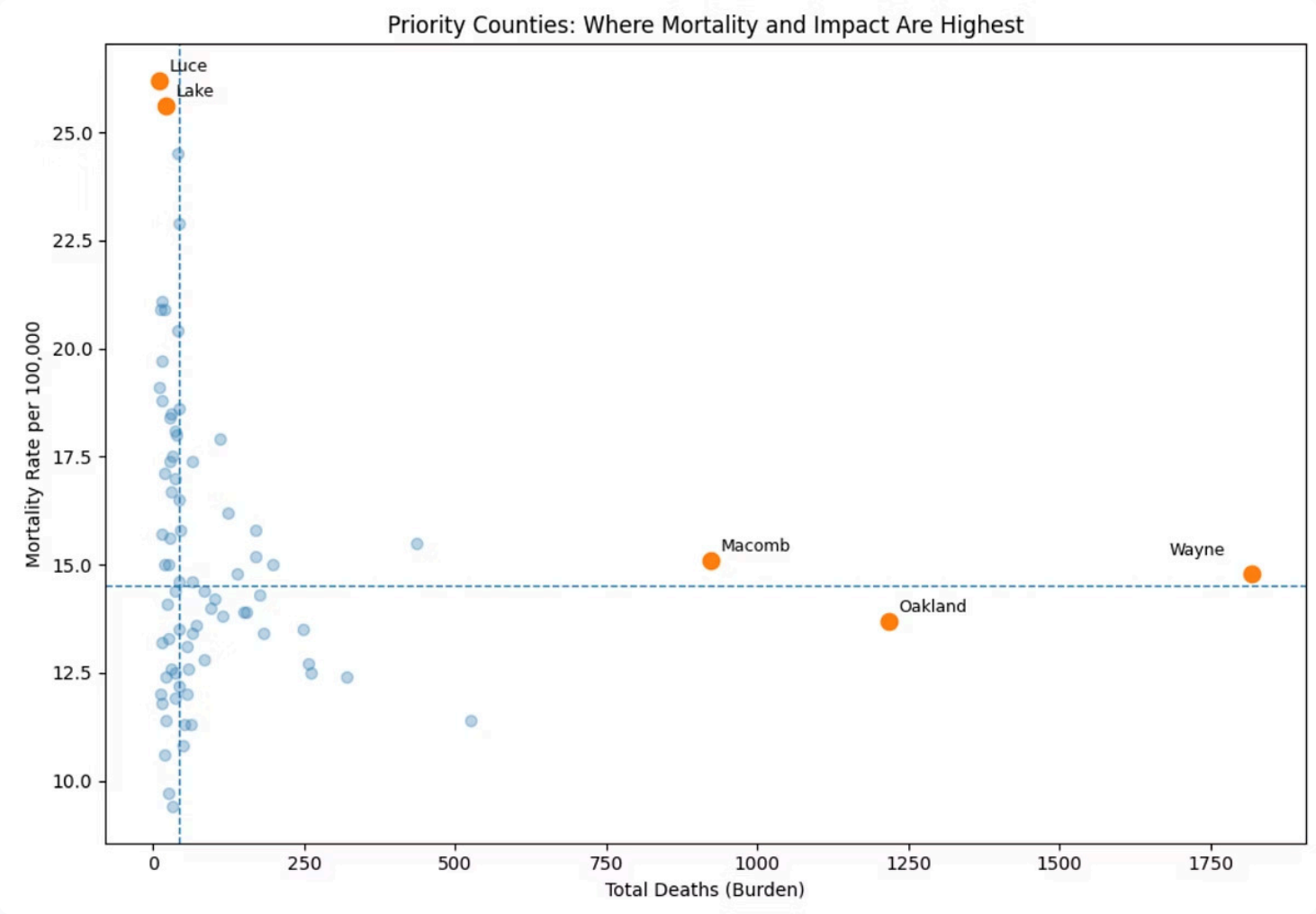

Where Mortality Is Most Concentrated

Five counties account for approximately 40% of total breast cancer deaths in Michigan, driving a disproportionate share of mortality burden.

Prioritization factors:

- Mortality rate (risk per capita)
- Total deaths (absolute impact)
- Not based on screening levels

40%

Total Deaths

Share attributed to top 5 counties

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Mortality Differences Are Driven by Compounding Risk Factors

Mortality is linked to broader risk factors that compound over time, creating disparities that screening alone cannot address.

How Risk Factors Compound

Higher Poverty & Food Insecurity

Lower income and food insecurity levels correlate strongly with elevated mortality rates

Older Populations

Counties with older demographics face higher baseline risk and delayed care-seeking behavior

Risk Factors Cluster

These factors concentrate in high-mortality counties, creating compounding effects

Differences in outcomes are associated with compounding risk factors, not screening rates

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What Should Be Done

Reducing mortality requires system-level improvements that focus beyond screening, specifically on what happens after detection.

Care Pathway After Screening

Where outcomes are determined

Improve Follow-Up

Ensure abnormal screenings lead to timely diagnosis by setting clear benchmarks, such as a 90-day follow-up standard

Strengthen Care Coordination

Standardize referral pathways and improve access to regional cancer centers for smoother transitions to treatment

Target High-Impact Counties

Focus resources where mortality burden is highest and use data to guide prioritization decisions

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Reducing Mortality Requires Targeted, System-Level Interventions

Disparities are multi-factor driven

Complex interplay of socioeconomic, demographic, and healthcare access factors

Screening alone is not sufficient

High screening rates do not guarantee improved survival outcomes

Targeted, system-level strategies are required

Data-driven interventions focused on high-mortality counties and care coordination

Reducing mortality requires focusing beyond screening and improving what happens after detection, especially in the communities driving the greatest burden.

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